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An engineer and researcher with a background in eVTOL conceptual design, aerodynamic shape optimization, and aero-structural analysis across academia and industry. Experienced in developing integrated multidisciplinary design optimization (MDO) frameworks to balance aerodynamic efficiency, propulsion, structural sizing, and weights. Dedicated to bridging advanced numerical simulation, optimization methods, and physical testing to support flight-ready eVTOL aircraft.

EDUCATION

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| 2024 - present | Ph.D. in Mechanical Engineering, HKUST, CGA : 4.30/4.00
Courses : Foundation of Solid Mechanics, Finite Element Methods, Data-Driven Modeling and Control of Dynamic Systems, Advanced Numerical Methods I.
Thesis : <i>eVTOL Aircraft Conceptual Design Optimization (tentative)</i>
Advisor : Prof. Rhea Liem |
| 2021 - 2023 | M.Eng. in Aerospace Engineering, Tohoku University, CGPA : 3.92/4.00 (Scholaro Link)
Courses : Mathematical Modeling and Computation, System Control Engineering, Numerical Analysis, Fluid Design Informatics, High Performance Computing, Fluid Dynamics, Structural Mechanics, etc.
Thesis : <i>Deep Learning Techniques for Aerodynamic Wing Shape Optimization</i>
Advisors : Prof. Shigeru Obayashi and Prof. Koji Shimoyama |
| 2017 - 2021 | B.Eng. in Mechanical and Aerospace Engineering, Tohoku University, CGPA : 3.96/4.00 (Scholaro Link)
Courses : Aircraft Design, Fluid Mechanics I and II, Theory of Elasticity, Computational Fluid Dynamics, etc.
Thesis : <i>An Artificial Neural Network-Assisted Genetic Algorithm with Application to Multi-Objective Transonic Airfoil Shape Optimization</i> . Advisor : Prof. Koji Shimoyama. |

EXPERIENCE

- | | |
|----------------------------------|---|
| February 2024 - present | PhD Researcher in Aircraft Design, HKUST, Hong Kong <ul style="list-style-type: none">➤ Achieved PhD candidacy after successfully passing the PhD Qualifying Examination, defending a research proposal on future-aware eVTOL design optimization and time-explicit LCA.➤ Developed a Hybrid MDF-IDF Multidisciplinary Design Optimization (MDO) framework in OpenMDAO to size and optimize Lift-Plus-Cruise and Multirotor eVTOLs, integrating aerodynamics, rotor performance, trim calculation, battery, motor, and weight estimation.➤ Formulated a multi-fidelity distributed iterative optimization architecture to coordinate system-level sizing with component-level detailed designs (e.g., VLM-based wing shape, BEMT-based rotor blade shape, battery pack topology optimization) via fixed-point physical parameter updates.➤ Pioneered a future-aware design methodology incorporating battery technology projections to determine cross-scenario robust designs and optimal discrete upgrade schedules using genetic algorithms.➤ Authored multiple research manuscripts detailing eVTOL MDO, technology forecasting, and system-component coordination, currently in preparation for submission to journals.➤ Maintained a perfect GPA (4.30/4.00) by achieving perfect scores (A+) in three advanced computational course projects : structural wing sizing using one- and two-way FSI, RL-based adaptive PID pitch control of Cessna-172, and a Roe-MUSCL finite-volume solver for 1D Euler equations. <div style="display: flex; gap: 5px;">OpenMDAOOpenAeroStructANSYSPythonMachine LearningeVTOL Aircraft Conceptual Design Optimization</div> |
| Feb - May 2025
Feb - May 2026 | Graduate Teaching Assistant MECH 3620 - Aircraft Design, HKUST, Hong Kong <ul style="list-style-type: none">➤ Facilitated interactive tutorial sessions using Julia (Pluto.jl), instructing students on computational methods and programming for aircraft conceptual design.➤ Taught comprehensive tutorial topics including : Overall Aircraft Design through the AeroFuse framework, Introduction to Julia Programming, Constraint Analysis (Matching Chart), Preliminary Weight Estimation, Low-to-Mid Fidelity Fixed-Wing Aerodynamics (DATCOM and Vortex Lattice Method), Mission and Trim Analysis, Weight & Balance, Stability Analysis, and Velocity-Loads (V-n) Diagrams.➤ Co-authored the final group design project definitions (Request for Proposal) with Prof. Zhenyu Gao, formulating detailed mission profiles, performance constraints, and certification requirements for a narrow-body commercial jet (2025) and a high-efficiency regional jet (2026).➤ Evaluated and graded student design projects : assessing midterm and final presentations, as well as comprehensive technical reports based on technical quality, trade-off analyses, and design rationales.➤ Provided continuous technical support and project consultation to student groups. <div style="display: flex; gap: 5px;">AeroFuseOpenVSPXFLR5XFOILJuliaPluto.jlLaTeX</div> |

August 2023 -
January 2024

Structural Analysis Engineer (full-time) | PT Turkish Aerospace Indonesia, BANDUNG, Indonesia

- Contributed as an analysis engineer for two projects : 1. Weight Reduction for ATA 27 Components (Flight Control System) of the TAI Hürjet (Advanced Jet Trainer), 2. Preliminary Design Study of the Horizontal Stabilizer for the TAI T625 Gökbey (Transport and Light Utility Helicopter).
- Performed static (SOL 101) and normal mode (SOL 103) analyses to determine structural integrity (Reserve Factor) using finite element methods (MSC Patran/Nastran) backed with hand calculations.
- Responsible for analyzing ATA 27 Components : 1) Rudder (Installation, Mounting Bracket, Lever Arm), 2) LEFAS Assy. (PDU Instl., Control Module Fitting, Bevel Gear), 3) Flaperon Instl., 4) IMU Instl.
- Performed hand calculation analyses for : 1) bolted joint, 2) lug and pin, 3) flanged stud.
- Prepared FEM : Load and BCs, 3D Mesh TET10, SPC, MPC (RBE2 vs RBE3), CBUSH, CGAP, etc.
- Automated the application of MPC and CBUSH using scripting in Patran Command Language.
- Performed aerodynamic analysis of the HorzStab using the Schrenk method, Vortex Lattice Method, and 3D Panel method to provide spanwise and chordwise load distribution.
- Calculated the shear force, bending moment, and torsion due to aerodynamic forces on the HorzStab.
- Wrote pyNastran code to extract forces/stresses automatically from HorzStab GFEM via OP2 files.
- Joined training about regulations : DOA (CS-21 Subpart J) and CS-29 for large helicopter.

MSC Patran Nastran Patran Command Language Hypermesh PyNastran OpenVSP Python MS Office

August 2022 -
July 2023

Aerodynamics Engineer (part-time) | teTra aviation corp., Tokyo, Japan

- Performed a multi-objective design exploration study of a tandem wing configuration to maximize aerodynamic performances and meet stability requirements for a forward flight of our eVTOL (Mk-5).
- Spearheaded the development of an aircraft-level integration environment (aerodynamics, structures, propulsion, weights, controls) using the SUAVE framework (from ADL Stanford University).
- Spearheaded the development of a tool for airfoil exploration and selection (1500+ airfoils database).
- Performed multi-fidelity subsonic aerodynamic analyses of eVTOL fixed-wing multi-rotor systems.
- Managed simulation portfolio (3D geometry, mesh, CFD data, automation pipeline, etc.)
- Performed conceptual design (eVTOL iterative sizing, weight breakdown analysis, required battery power over an intended flight profile, reversed engineering process, what-if analyses, etc.)
- Assisted manned flight test preparation at Fukushima RTF (operation, flight log analysis, PID tuning).

Python Linux OpenVSP Xflr5 XFOIL ANSYS Fluent SUAVE Flight Review (PX4) Notion

April 2020 -
September 2023

Student Researcher | Fluids Engineering with Data Science Laboratory, TOHOKU UNIVERSITY, Japan

- Worked on collaborative research projects under “The Program for Promoting Research on the Super-computer Fugaku” by MEXT and “Boeing Higher Education Program” by the Boeing Japan.
- Developed an in-house surrogate-based optimization framework that utilizes a dynamically retrained multilayer perceptron combined with a genetic optimizer.
- Applied the framework to aerodynamic design optimization of 2D and 3D transonic wings : PARSEC, B-spline airfoils, and NASA Common Research Model (CRM) wing.
- Developed a DCGAN-based generative method to produce synthetic wing designs and a CNN-based geometric filtering method to filter abnormal shapes and efficiently explore the design space.
- Collaborated with researchers at IFS to apply the framework to wing structural layout optimization.
- Automated the geometry production, meshing, CFD analysis, and optimization on an HPC system.
- Performed wind tunnel experiments on a NACA 4415 airfoil, measuring lift/drag coefficients and wake velocity distributions across angles of attack, including stall onset characterization.
- Investigated laminar-to-turbulent flow transition in pipe flow, measuring intermittency factors and turbulent patch growth across Reynolds numbers.

Python C Linux HPC SU2 Pointwise MACH-Aero Tecplot PyTorch Pymoo Wind Tunnel

SKILLS

Aircraft Conceptual Design	OpenVSP, Flow5, Xflr5, XFOIL, XROTOR, SUAVE, AeroFuse
3D Geometry and Visualization	OpenVSP, SolidWorks, FreeCAD, ANSYS DesignModeler, PyGeo, Tecplot, Paraview
CFD and Meshing	ANSYS Fluent, ADflow, SU2, Pointwise, OpenAeroStruct
Finite Element Analysis	MSC Patran, Nastran, Hypermesh, PyNastran, ANSYS Mechanical
Multidisciplinary Design Optimization	OpenMDAO, ANSYS optiSLang, Pymoo
Experimental Aerodynamics	Basic Wind Tunnel Testing (airfoil CL/CD, velocity and stall characterization)
Flight Control and Analysis	Knowledgeable (Flight Review); Informed (PX4, jMAVSim, QGroundControl)
Programming Languages	Proficient (Python, C, Matlab/Simulink); Knowledgeable (C++, HTML5, CSS3, JavaScript)
ML and Data Science	NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, TensorFlow, Spreadsheet
Operating System and SCM	MacOS, Microsoft Windows, Linux Ubuntu; Git, GitHub, GitLab, Sublime Merge
Text Editing and Documentation	Sublime Text, VIM, Jupyter Notebook, Microsoft Word, Notion, LaTeX Overleaf
Language Proficiency	Professional (English : iBT 107/120); Conversational (Japanese N3); Native (Indonesian)

PROJECTS

MCEVS : MULTI-CONFIGURATIONAL eVTOL SIZING

JUNE 2024 - PRESENT

 [GitHub Repo](#) : sizing and optimization framework for eVTOL configurations (PhD research project)

TERM PROJECT : ONE- AND TWO-WAY FLUID STRUCTURE INTERACTION IN WING STRUCTURAL SIZING

SEPT - DEC 2025

 Final Report : CFD-FEA wing sizing combined with a surrogate-based optimization workflow

TERM PROJECT : AIRCRAFT PITCH CONTROL VIA REINFORCEMENT LEARNING-BASED ADAPTIVE PID

SEPT - DEC 2024

 [GitHub Repo](#) : Cessna-172 flight simulation + automatic pitch controller

WORK PACKAGE : WEIGHT REDUCTION FOR ATA27 COMPONENTS OF THE TAI HÜRJET

AUG 2023 - JAN 2024

 PT Turkish Aerospace Indonesia : structural analysis engineer

DEEP LEARNING TECHNIQUES FOR AERODYNAMIC WING SHAPE OPTIMIZATION

2021 - 2022

 Boeing Higher Education organized by Prof. Shigeru Obayashi and reported to the President of Boeing Japan.

LEADING RESEARCH ON INNOVATIVE AIRCRAFT DESIGN TECHNOLOGIES TO REPLACE FLIGHT TEST

2020 - 2022

 Fugaku Supercomputer Project : AI applications in aircraft design (led by Prof. Koji Shimoyama)

PUBLICATIONS

Articles

- > Hariansyah, M. A., Reyner, C., and Liem, R. P., "Mapping Optimal eVTOL Configuration Selection Across Missions and Battery Technology Timeline," *under preparation for journal submission*.
- > Hariansyah, M. A., and Shimoyama, K., "An Artificial Neural Network-Assisted Genetic Algorithm With Application to Multi-Objective Transonic Airfoil Shape Optimization," *JAXA Special Publication : Proceedings of the 53rd Fluid Dynamics Conference/39th Aerospace Numerical Simulation Symposium*, 2022, pp. 115-124, JAXA-SP-21-008, ISSN 2433-2232.
- > Hariansyah, M. A., and Shimoyama, K., "On the Use of a Multilayer Perceptron Based Surrogate Model in Evolutionary Optimization," *Proceedings of the Computational Mechanics Conference*, 2021, Vol. 2021.34, Online ISSN 2424-2799, DOI:10.1299/jsmecmd.2021.34.235

Presentations

- > Hariansyah, M. A., Reyner, C., Liem, R. P., and Palar, P. S., "Optimal Conceptual Design and Upgrade Strategies for eVTOL Aircraft Under Evolving Technologies," *35th Congress of the ICAS*, Sept 2026, Sydney, Australia, DOI:10.71945/icas2026_0631
- > Liu, L., Wu, Y., Hariansyah, M. A., and Liem, R. P., "A Bilevel Strategy for On-Design eVTOL Sizing and Off-Design Structural Weight Optimization," *35th Congress of the ICAS*, Sept 2026, Sydney, Australia, DOI:10.71945/icas2026_0209
- > Hariansyah, M. A., and Liem, R. P., "A Physics-Driven Multifidelity Distributed Iterative Optimization Framework for eVTOL Aircraft Design," *14th ASMO UK / 3rd ASMO-EUROPE / ISSMO Conference*, Jul 2026, London, United Kingdom.
- > Hariansyah, M. A., and Shimoyama, K., "Deep Learning Techniques for High-Dimensional Surrogate-Based Aerodynamic Design," *33rd Congress of the ICAS*, Sept 2022, Stockholm, Sweden.
- > Hariansyah, M. A., and Shimoyama, K., "Aerodynamic Wing Shape Optimization via Deep Learning-Assisted Genetic Algorithm," *JSME Annual Meeting 2022*, Sept 2022, Toyama, Japan.
- > Inaba, Y., Date, S., Hariansyah, M. A., Abe, Y., Shimoyama, K., Okabe, T., and Obayashi, S., "Optimization of Structural Layout for Composite Aircraft Wings," *the 18th ICFD*, Oct 2021, Sendai, Japan.
- > Hariansyah, M. A., and Shimoyama, K., "On the Use of a Multilayer Perceptron as an Aerodynamic Performance Approximator in Multi-Objective Transonic Airfoil Shape Optimization," *the 18th ICFD*, 2021, Sendai, Japan.

HONORS & AWARDS

January 2024	Hong Kong PhD Fellowship Scheme (prestigious program from the University Grants Committee of HK)
June 2022	IFS Graduate Student Overseas Presentation Award (a travel grant 350k JPY)
2021 - 2023	Mizuho International Foundation Scholarship Awardee (Top 15/ 60+ international applicants).
2021 - 2022	Boeing Higher Education (BHE) Program Student Research Project Awardee. (JPY 300k grant)
October 2021	Best Presentation Award at the 18th International Conference on Flow Dynamics (of 71 papers).
2017 - 2021	Japanese Government (MEXT) Scholarship Awardee (Top 10/100+ applicants globally).

CERTIFICATES & TRAINING

August 2025	"Design and Simulate the Aerodynamics of Propellers" by Udemy (4 hours)
March 2025	"Electric VTOL Aircraft Design : Theory and Practice" by AIAA/VFS (20 Hours)
December 2024	"Flight Mechanics - From Theory to Certification of Aircraft" by Udemy (10 hours)
October 2023	"Design Organization Approval Training (CS-21 Subpart J)" by PT Turkish Aerospace Indonesia
July 2023	"Japanese Language Proficiency Test (JLPT) N3" by the Japan Foundation